

An All-Face Affine and Laguerre Theorem for the CIR Square-Root Diffusion

HCS Research Program

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Abstract

We close one theorem across the full nonnegative parameter cone of the Cox–Ingersoll–Ross square-root diffusion. The scale/speed test separates the entrance face $2\kappa\theta \geq \sigma^2$ from the regular, instantaneously reflecting face, while zero-dimensional and zero-noise faces are treated as distinct degenerations. A single affine Riccati flow gives all Laplace transforms and, in the interior, the noncentral chi-square transition law. The invariant Gamma law puts the generator in Laguerre form: the exact reversible kernel, sharp gap κ , and L^2 /chi-square/TV contraction follow from one orthogonal expansion. This is a stochastic Markov-semigroup result, not a primitive-orbit zeta or a Hilbert–Pólya construction. Independent exact reconstruction (235 assertions), symbolic identities (18), byte replay and 20 hostile mutations audit the certificate.

1 Object and boundary atlas

We freeze, in physical time $t \geq 0$,

$$dX_t = \kappa(\theta - X_t) dt + \sigma\sqrt{X_t} dW_t, \quad X_0 = x \geq 0, \quad \kappa, \theta, \sigma \geq 0. \quad (1)$$

The square-root modulus gives a unique nonnegative strong solution. The interior generator is $\mathcal{L}f = \kappa(\theta - x)f' + (\sigma^2/2)xf''$.

Theorem 1 (Feller and degenerate-face classification). *Assume $\kappa, \theta, \sigma > 0$ and write $\alpha = 2\kappa\theta/\sigma^2$. The origin is an entrance (hence inaccessible from $x > 0$) iff $\alpha \geq 1$, including equality $2\kappa\theta = \sigma^2$. If $0 < \alpha < 1$, it is regular and the canonical CIR solution is instantaneously reflecting. If $\theta = 0$ or $\kappa = 0$ with $\sigma > 0$, the dimension is zero and 0 is absorbing. If $\sigma = 0$ and $\kappa > 0$, the path is deterministic, $X_t = \theta + (x - \theta)e^{-\kappa t}$; when also $\theta = 0$ it decays toward the absorbing origin. If $\kappa = \sigma = 0$, every starting point is constant.*

Proof. Near zero the scale density is proportional to $x^{-\alpha}$ and the speed density to $x^{\alpha-1}$. The scale integral diverges exactly for $\alpha \geq 1$, while the speed integral is finite for every $\alpha > 0$; Feller’s test gives entrance versus regular. At dimension zero the SDE has no inward drift at the boundary and pathwise uniqueness makes it absorbing. Setting $\sigma = 0$ reduces (1) to the displayed ODE, which also resolves the remaining faces. $\square$

2 Affine transform and transition law

For $u \geq 0$, use $\mathbb{E}_x e^{-uX_t} = \exp[-\phi_t(u) - \psi_t(u)x]$. Substitution in the backward equation gives

$$\psi' = -\kappa\psi - \frac{1}{2}\sigma^2\psi^2, \quad \phi' = \kappa\theta\psi, \quad \psi(0) = u, \quad \phi(0) = 0. \quad (2)$$

For $\kappa, \sigma > 0$, put $h_t = \sigma^2(1 - e^{-\kappa t})/(2\kappa)$; then

$$\psi_t = \frac{ue^{-\kappa t}}{1 + h_t u}, \quad \phi_t = \frac{2\kappa\theta}{\sigma^2} \log(1 + h_t u). \quad (3)$$

The continuous faces are $\psi = u/(1 + \sigma^2 tu/2)$, $\phi = 0$ for $\kappa = 0$, and $\psi = ue^{-\kappa t}$, $\phi = \theta u(1 - e^{-\kappa t})$ for $\sigma = 0$; with both rates zero, $(\phi, \psi) = (0, u)$.

Equation (3) is equivalently

$$X_t = c_t \chi_{4\kappa\theta/\sigma^2}^2(\lambda_t), \quad c_t = \frac{\sigma^2(1 - e^{-\kappa t})}{4\kappa}, \quad \lambda_t = \frac{4\kappa e^{-\kappa t} x}{\sigma^2(1 - e^{-\kappa t})}. \quad (4)$$

For dimension zero, (4) is interpreted with its atom at zero; for example when $\theta = 0$, $\mathbb{P}_x(X_t = 0) = \exp[-xe^{-\kappa t}/h_t]$.

3 Gamma law, Laguerre kernel and gap

On the positive interior set $\beta = \sigma^2/(2\kappa)$. The density

$$\pi(x) = \frac{x^{\alpha-1} e^{-x/\beta}}{\Gamma(\alpha) \beta^\alpha} \quad (5)$$

is invariant and unique; its mean and variance are θ and $\theta\sigma^2/(2\kappa)$. With $z = x/\beta$,

$$\mathcal{L} = \kappa\{z\partial_z^2 + (\alpha - z)\partial_z\}. \quad (6)$$

The Laguerre equation and orthogonality are

$$zL_n'' + (\alpha - z)L_n' + nL_n = 0, \quad \int_0^\infty L_n L_m z^{\alpha-1} e^{-z} dz = \frac{\Gamma(n+\alpha)}{n!} \delta_{nm}.$$

Consequently, for $t > 0$ the exact reversible transition kernel is

$$p_t(x, y) = \pi(y) K_t(x, y), \quad K_t = \sum_{n \geq 0} e^{-\kappa n t} \frac{n! \Gamma(\alpha)}{\Gamma(n+\alpha)} L_n^{(\alpha-1)}(x/\beta) L_n^{(\alpha-1)}(y/\beta). \quad (7)$$

Proposition 1 (Sharp mixing). *For $f \in L^2(\pi)$,*

$$\text{Var}_\pi(P_t f) \leq e^{-2\kappa t} \text{Var}_\pi(f).$$

If $d\mu/d\pi \in L^2(\pi)$, then $\chi^2(\mu P_t || \pi) \leq e^{-2\kappa t} \chi^2(\mu || \pi)$ and $\|\mu P_t - \pi\|_{\text{TV}} \leq \frac{1}{2} e^{-\kappa t} \sqrt{\chi^2(\mu || \pi)}$. The first Laguerre mode attains equality, so the gap is exactly κ .

Proof. Equation (6) is symmetric in $L^2(\pi)$ and (7) diagonalizes its closure. All nonconstant modes have eigenvalues at most $-\kappa$; Parseval gives the variance and chi-square bounds, and Cauchy–Schwarz gives TV. The mode $L_1^{(\alpha-1)} = \alpha - z$ has eigenvalue $-\kappa$. $\square$

References

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- [2] W. Feller, “Two singular diffusion problems,” *Annals of Mathematics* 54 (1951), 173–182. DOI.
- [3] A. Alfonsi, “High order discretization schemes for the CIR process,” *Monte Carlo Methods and Applications* 16 (2010), 1–28. DOI.

Declarations

Scope. NO_BAD_EULER_OR_ROOT_NUMBER; no target arithmetic, Euler-factor, target-zero, automorphy, or Hilbert–Pólya claim. **Data and code.** All exact formulas, ledgers and audit programs accompany HCS-C229. **AI-use disclosure.** Generative tools assisted drafting and code generation; internal checks validate the displayed identities. This is not external peer review.